

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Project ID: 25-26J-090

Project Title:

**AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT
MANAGEMENT IN STREAMING ENVIRONMENTS**

1. Introduction

1.1. Background:

Modern organizations increasingly rely on real-time data pipelines to ingest, process, and analyze high-velocity data from diverse sources such as IoT sensors, applications, logs, and user interactions. These streaming environments enable timely insights and data-driven decision-making but also introduce significant challenges related to data quality, privacy, and system reliability.

As data flows continuously, it is often affected by quality issues such as missing values, anomalies, duplicates, and inconsistencies. At the same time, sensitive information such as personally identifiable information (PII) may be exposed if privacy controls are not enforced early in the pipeline. Additionally, evolving data sources frequently introduce schema drifts, which can silently break pipelines or corrupt downstream analytics. Unstructured data streams further complicate this landscape due to their lack of predefined structure and scalability requirements.

Given these challenges, there is a growing need for an autonomous, real-time framework capable of ensuring data quality, preserving privacy, managing schema evolution, and handling both structured and unstructured data streams without manual intervention.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

1.2. Research Problem:

Despite the critical role of streaming data pipelines, existing solutions largely address data quality, privacy, unstructured data processing, and schema management in isolation. Many systems rely on post-ingestion validation, manual monitoring, or static rules, which are insufficient for high-velocity and continuously evolving data environments.

The lack of integrated, real-time, and autonomous mechanisms results in delayed error detection, silent data corruption, privacy violations, regulatory non-compliance, and increased operational overhead. Schema drift often remains undetected until downstream failures occur, while unstructured data quality assessment lacks scalable automation. Consequently, organizations struggle to maintain reliable, secure, and trustworthy data pipelines. This research addresses the need for a unified autonomous framework that simultaneously manages data quality, privacy, schema drift, and unstructured data preprocessing in streaming environments.

1.3. Objectives:

It Number	Objective	Objective number
IT22893666	To design a real-time data quality assessment and explainable preprocessing framework with benchmarking and rollback support for structured data streams	1
IT22556684	To develop an automated real-time privacy preservation mechanism that detects, classifies, and protects sensitive data in streaming pipelines	2
IT22549136	To design an autonomous multi-agent framework for real-time quality assessment and preprocessing of unstructured data streams ensuring scalability and reliability	3
IT22204752	To implement an automated schema drift detection and management mechanism that adapts to evolving data schemas without disrupting streaming pipelines	4

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

2. Data Exploration

2.1. Data Collection

This research uses a combination of Verified synthetic datasets and public Kaggle datasets to evaluate the proposed autonomous framework in streaming environments.

For the real-time data quality assessment and explainable preprocessing framework for structured data, the research does not depend on a fixed dataset; instead, public Kaggle structured datasets are dynamically ingested to evaluate quality detection, correction, benchmarking, and rollback mechanisms in real-time, ensuring adaptability to varying data characteristics. All the testing datasets are verified by industry experts to achieve that these data are the same structure as the industry datasets. At the same time business validation rules are collected through a hybrid human-AI pipeline where domain expert documents (PDF/TXT/DOCX formats) are processed using Groq LLM to automatically extract structured constraints.

The dataset used in this research was constructed by integrating data from multiple sources. Relevant attributes were extracted from selected columns of an existing dataset, supplemented with data obtained through web scraping, and further enhanced through synthetically generated data. The synthetic dataset was created to train a small language model (SLM) and was used for both training and testing purposes. Synthetic data generation was guided by standard data protection and privacy regulations, including GDPR, PDPA, and PCI, to ensure regulatory alignment and avoid the use of real sensitive information. All data sources were unified into a single dataset to ensure consistency and completeness. Each data instance was stored in JSON format, with textual content annotated using the BIO (Begin – Inside - Outside) tagging scheme to support structured sequence labeling and downstream analysis.

The proposed Agentic AI framework employs verified synthetic unstructured text data during the model training phase to enable controlled learning of autonomous agent behaviors under diverse data quality conditions, including noise, corruption, language variability, and semantic inconsistencies. This synthetic data supports the training of specialized text-processing agents such as language detection, text normalization, tokenization, validation, and semantic integrity agents, ensuring robust and reproducible agent coordination. For evaluation and testing, the framework utilizes publicly available unstructured text and log-based datasets from Kaggle, representing realistic real-world data distributions and quality challenges. Additionally, synthetic streaming datasets are used to simulate high-velocity, irregular, and corrupted data streams, enabling controlled assessment of real-time multi-agent scalability, coordination efficiency, and end-to-end data integrity in the absence of readily available real-world streaming datasets.

For the schema drift detection and management component, the system was built and configured using a structured, scenario-based synthetic dataset designed to explicitly model schema expectations and deviations. This dataset consists of paired *expected* and *observed* data files, supported by schema mapping definitions and metadata specifications. The *expected* datasets represent the canonical or reference schema, while the *observed* datasets simulate incoming data streams. Metadata and mapping files were used to define schema

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

rules, field-level correspondences, and validation constraints. These artifacts enabled the construction of core system logic, including schema comparison, drift classification, rule-based decision-making, and validation workflows, without relying on machine learning model training. To evaluate the effectiveness of the system, versioned synthetic datasets were used to simulate controlled schema evolution across multiple data versions. These versions include common real-world schema drift scenarios such as column addition, removal, renaming, and data type or nullability changes. By applying the detection framework to these evolving datasets, the system's ability to identify, classify, and manage schema drift in real time was systematically assessed. This clear separation between system construction datasets and testing datasets ensures unbiased evaluation and demonstrates the robustness of the proposed approach under reproducible and realistic streaming conditions.

All selected datasets and preprocessing outcomes are validated by industry experts to ensure practical relevance and correctness.

Feedback from experts:



Dhanushka NADEESHAN

to me ▾

15:41 (7 hours ago) ☆ 😊 ↩ ⋮

Dear Amanda

I have gone through the attached dataset and its intended use within your research on data privacy in data pipelines. The dataset appears to be reasonably structured and appropriate for academic and experimental purposes at this stage.

While the dataset may not fully capture all real-world complexities found in large-scale industrial systems, it is suitable for demonstrating concepts, experimentation, and evaluation within an academic research context. Minor refinements may be possible depending on the specific scope of analysis, but no critical issues were identified that would prevent its use.

Based on my review, I confirm that the dataset can be used for your stated research objectives.

Kind Regards,

Dhanushka NADEESHAN Mr. | Software Developer

Office: +94 112687868

www.megatrendgroup.com

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Dear Amodhya,

This email is to confirm that the dataset and test cases used for training and evaluating the Agentic AI model for relevant text selection in LLM training have been reviewed and validated.

The dataset conforms to industry-standard practices for unstructured text processing and includes representative real-world quality issues. The associated test cases adequately assess agent selection, preprocessing accuracy, and robustness, and are suitable for evaluating autonomous agentic workflows.

Based on the review, the dataset and test cases are considered appropriate and reliable for academic research and industry-aligned experimentation.

--

Thanks & Regards,

NEXT

Tharusha Kasthuriarachchi
Data Analyst
Business Intelligence Systems
Email: tharusha_kasthuriarachchi@next.co.uk
Tel: +94761133592

DISCLAIMER

This email is confidential and subject to important disclaimers and conditions in relation to monitoring, viruses, confidentiality and legal privilege full details of which can be viewed on our Email Policy at the following link: <http://www.next.co.uk/privacy/>

Next Holdings Ltd registered in England 35161. Registered Office: Deodar Road, Enderby Leicester LE19 4AT. Next Retail Ltd is authorised and regulated by the Financial Conduct Authority for Consumer Credit.

Disclaimer: This email and any attachment transmitted herewith are confidential and is intended solely for the use of the individual or entity to which they are addressed and may contain information that is privileged or otherwise protected from disclosure. If you are not the intended recipient, you are hereby notified that disclosing, copying, distributing, or taking any action in reliance on the contents of this information is strictly prohibited. If you have received this email in error, please notify the sender immediately by reply email and discard all of its contents by deleting this email and the attachment, if any, from your system.

Dear Nanduni,

Thank you for providing the dataset for evaluation.

I have examined the dataset in the context of your research on schema drift management in data pipelines. The dataset is well-prepared and aligned with the objectives of your study, and it adequately represents schema changes commonly observed in real-world data systems.

From an industry viewpoint, this dataset is appropriate for experimental validation and can be confidently used to support your academic research. No major revisions are necessary at this stage.

I appreciate the diligence shown in developing this dataset and wish you the very best in the successful completion of your research.

Kind Regards,

Dhanushka NADEESHAN Mr. | Software Developer
Office: +94 112687868
www.megatrendgroup.com

Dear Lavindi,

This email is to confirm that the structured dataset used for the structured data model, along with the defined test scenario datasets, has been reviewed and evaluated.

The dataset and test scenarios meet industry standards in terms of data quality, consistency, coverage, and relevance. They are appropriate for validating structured data processing, modeling accuracy, and real-world applicability within an industry-aligned environment.

Based on the review, the datasets are considered suitable and reliable for academic research and industry-standard experimentation.

--

Thanks & Regards,

NEXT

Tharusha Kasthuriarachchi
Data Analyst
Business Intelligence Systems
Email: tharusha_kasthuriarachchi@next.co.uk
Tel: +94761133592

DISCLAIMER

This email is confidential and subject to important disclaimers and conditions in relation to monitoring, viruses, confidentiality and legal privilege full details of which can be viewed on our Email Policy at the following link: <http://www.next.co.uk/privacy/>

Next Holdings Ltd registered in England 35161. Registered Office: Deodar Road, Enderby Leicester LE19 4AT. Next Retail Ltd is authorised and regulated by the Financial Conduct Authority for Consumer Credit.

Disclaimer: This email and any attachment transmitted herewith are confidential and is intended solely for the use of the individual or entity to which they are addressed and may contain information that is privileged or otherwise protected from disclosure. If you are not the intended recipient, you are hereby notified that disclosing, copying, distributing, or taking any action in reliance on the contents of this information is strictly prohibited. If you have received this email in error, please notify the sender immediately by reply email and discard all of its contents by deleting this email and the attachment, if any, from your system.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

3. Dataset Description:

NO	Data source	Description	Resource	Size	Key attributes
1	Synthetic (Structured Streaming)	Generated sensor/transaction-like streams with injected missing values, duplicates, outliers/anomalies for real-time quality assessment, explainable preprocessing, benchmarking, and rollback evaluation	Locally generated (custom)	Configurable (e.g., 10k–1M rows depending on experiments)	Depends on the dataset
2	Synthetic (for NER model)	A synthetic dataset was generated to train the NER model, simulating realistic sensitive and non-sensitive entities without using real personal data. All text instances were annotated using the BIO tagging scheme to support supervised training and evaluation.	Locally generated (custom)	10000	Data in json format (value – label pairs with BIO tagging)
3	Synthetic (for SLM)	The regulatory dataset, based on GDPR, PDPA, and PCI, was structured in a Q&A format and stored in JSON to ensure	Generated	400-500	Data in json format (QnA style)

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

		consistency, readability, and easy integration for analysis.			
4	Synthetic Unstructured Text Dataset	Artificially generated text samples with controlled quality issues used to train the LLM to autonomously select appropriate preprocessing agents for each input	Internally generated synthetic data	50,000 text records	Language variability, spelling errors, grammatical noise, duplicated content, missing tokens, mixed-language text, malformed sentences
5	Synthetic (Schema Drift Versions)	Scenario-based synthetic datasets containing an expected (baseline) schema and an observed (drifted) schema to evaluate rename detection	Locally generated (custom)	160 scenarios 2512(Supervised pairs file: training_pairs.csv = 2,512 pairs)	Paired schema versions per scenario: expected.csv and observed.csv Ground-truth rename mapping: mapping.json Scenario drift metadata: meta.json + scenarios_meta.json Drift types included: column renames, added , dropped, type drifts, nullability shifts

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

6	Kaggle datasets (schema drifts)	Public Kaggle datasets were modified by adding, removing, and renaming columns to simulate schema drift for evaluation.	Kaggle (public dataset)	Varies by dataset	Depends on the data set since the system works for any data set
7	Kaggle (Structured Dataset)	Real-world structured dataset used to validate the framework under realistic noise and attribute diversity	Kaggle (public dataset)	Varies by dataset	mixed numeric/categorical fields, missing values, inconsistent formatting
8	Kaggle (Unstructured Dataset)	Real-world unstructured dataset (e.g., text/log-like data) used to test multi-agent preprocessing, quality scoring, and integrity checks in streaming simulation	Kaggle (public dataset)	Varies by dataset	Depends on the data set since the system works for any data set

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Snapshots of the datasets:

Schema drift renames detection model train dataset:

Expected.csv

	order_id	customer_id	order_date	product_id	quantity	unit_price	total_price	currency	payment_status
1	ORD_004172	CUST_004444	2024-02-19 08:22:42	PROD_003238	9	64.99	584.91	USD	PAID
2	ORD_004173	CUST_004445	2024-12-10 20:43:31	PROD_003239	1	24.6	24.6	EUR	PAID
3	ORD_004174	CUST_004446	2023-12-04 12:25:57	PROD_003240	8	38.39	261.05	GBP	PAID
4	ORD_004175	CUST_004447	2023-09-22 18:34:58	PROD_003241	4	6.44	25.76	USD	PAID
5	ORD_004176	CUST_004448	2023-11-30 04:53:37	PROD_003242	2	35.92	61.06	USD	PAID
6	ORD_004177	CUST_004449	2023-12-13 11:36:59	PROD_003243	3	16.38	49.14	USD	PAID
7	ORD_004178	CUST_004450	2023-12-19 08:26:09	PROD_003244	3	29.64	88.92	USD	FAILED
8	ORD_004179	CUST_004451	2024-04-25 17:10:18	PROD_003245	1	30.8	30.8	USD	PAID
9	ORD_004180	CUST_004452	2024-01-29 20:33:40	PROD_003246	6	34.78	208.68	LKR	PAID
10	ORD_004181	CUST_004453	2024-04-08 23:07:19	PROD_003247	1	16.1	16.1	EUR	PAID
11	ORD_004182	CUST_004454	2024-07-31 21:47:38	PROD_003248	6	36.54	208.28	LKR	PAID
12	ORD_004183	CUST_004455	2023-06-30 12:47:15	PROD_003249	1	28.87	28.87	EUR	PAID
13	ORD_004184	CUST_004456	2024-02-05 20:59:51	PROD_003250	3	26.36	79.08	USD	PENDING
14	ORD_004185	CUST_004457	2024-11-30 18:37:46	PROD_003251	7	99.12	693.84	GBP	PAID
15	ORD_004186	CUST_004458	2023-07-04 08:46:28	PROD_003252	8	19.83	150.71	GBP	PAID
16	ORD_004187	CUST_004459	2024-05-21 15:30:43	PROD_003253	5	12.44	59.09	LKR	PENDING
17	ORD_004188	CUST_004460	2023-09-05 07:31:58	PROD_003254	6	21.26	127.56	USD	PAID
18	ORD_004189	CUST_004461	2024-05-31 16:10:47	PROD_003255	4	58.69	234.76	GBP	REFUNDED

Observed.CSV

	order_id	customer_id	order_date	product_id_unit	quantity	price_unit	total	c	payment_status	shipping_cost	discount_rate	coupon_code
1	ORD_004172	CUST_004444	2024-02-19 08:22:42	PROD_003238	9	64.99	584.91		PAID	2.84	0.0	
2	ORD_004173	CUST_004445		PROD_003239	1	24.6	24.6	EUR	PAID	3.24	0.05	FREESHIP
3	ORD_004174	CUST_004446	2023-12-04 12:25:57	PROD_003240	8	38.39	261.05	GBP	PAID	5.94	0.1	
4	ORD_004175	CUST_004447		PROD_003241	4	6.44	25.76		PAID	0.79	0.15	NEW10
5	ORD_004176	CUST_004448		PROD_003242	2	35.92	61.06	USD	PAID	1.62	0.0	
6	ORD_004177	CUST_004449		PROD_003243	3	16.38	49.14		PAID	2.27		
7	ORD_004178	CUST_004450	2023-12-19 08:26:09	PROD_003244	3	29.64	88.92	USD	FAILED	0.58	0.0	
8	ORD_004179	CUST_004451	2024-04-25 17:10:18	PROD_003245	1	30.8	30.8	USD	PAID	1.66	0.1	NEW10
9	ORD_004180	CUST_004452	2024-01-29 20:33:40	PROD_003246	6	34.78	208.68	LKR	PAID	5.59	0.0	NEW10
10	ORD_004181	CUST_004453	2024-04-08 23:07:19	PROD_003247	1	16.1	16.1	EUR	PAID	2.38		
11	ORD_004182	CUST_004454	2024-07-31 21:47:38	PROD_003248	6	36.54	208.28	LKR	PAID	1.62		
12	ORD_004183	CUST_004455	2023-06-30 12:47:15	PROD_003249	1	28.87	28.87		PAID	2.08	0.0	
13	ORD_004184	CUST_004456	2024-02-05 20:59:51	PROD_003250	3	26.36	79.08	USD	PENDING	0.11	0.1	
14	ORD_004185	CUST_004457	2024-11-30 18:37:46	PROD_003251	7	99.12	693.84		PAID	2.38	0.05	
15	ORD_004186	CUST_004458	2023-07-04 08:46:28	PROD_003252	8	19.83	150.71	GBP	PAID	1.83	0.1	
16	ORD_004187	CUST_004459	2024-05-21 15:30:43	PROD_003253	5	12.44	59.09	LKR	PENDING	2.45	0.0	NEW10
17	ORD_004188	CUST_004460	2023-09-05 07:31:58	PROD_003254	6	21.26	127.56	USD	PAID	1.64	0.0	
18	ORD_004189	CUST_004461	2024-05-31 16:10:47	PROD_003255	4	58.69	234.76	GBP	REFUNDED	10.15	0.0	NEW10
19	ORD_004190	CUST_004462	2024-08-13 07:06:07	PROD_003256	9	13.32	101.9	USD	PAID	0.78	0.0	FREESHIP
20	ORD_004191	CUST_004463	2024-03-24 20:15:43	PROD_003257	9	119.26	1073.34	LKR	PAID	16.02	0.0	

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Paired.csv

	scenario_id	domain	old_col	new_col	label	pair_category
1	0	retail	customer_id	customer_id	1	typo
2	0	retail	product_id	product_id_unit	1	unit_variant
3	0	retail	currency	c	1	abbrev
4	0	retail	total_price	total	1	abbrev
5	0	retail	unit_price	price_unit	1	token_swap
6	0	retail	unit_price	total	0	negative_random
7	0	retail	product_id	customer_id	0	hard_group_trap
8	0	retail	unit_price	shipping_cost	0	hard_group_trap
9	0	retail	unit_price	c	0	negative_random
10	0	retail	unit_price	total	0	hard_group_trap
11	0	retail	customer_id	product_id_unit	0	hard_group_trap
12	0	retail	customer_id	total	0	negative_random
13	0	retail	total_price	price_unit	0	hard_group_trap
14	0	retail	total_price	product_id_unit	0	negative_random
15	0	retail	total_price	shipping_cost	0	hard_group_trap
16	1	sensor	sensor_status	v2_sensor_status	1	prefix_suffix
17	1	sensor	temperature_c	temp_celsius	1	synonym
18	1	sensor	sensor_id	si	1	abbrev

Unstructured dataset:

Go until ju long point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...
Ok lar... Joking wif u oni...
Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
U dun say so early hor... U c already then say...
Nah I don't think he goes to usf, he lives around here though
FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, â€1.50 to rcv
Even my brother is not like to speak with me. They treat me like aids patient.
As per your request 'Melle Melle (Oru Minnaminunginte Nurrungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
WINNER!! As a valued network customer you have been selected to receive a â€900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030
I'm gonna be home soon and i don't want to talk about this stuff anymore tonight, k? I've cried enough today.
SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH11 and send to 87575. Cost 150p/day, 6days, 16+ TsandCs apply Reply HL 4 info
URGENT! You have won a 1 week FREE membership in our â€100,000 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C www.dbuk.net LCCLTD POBOX 4403LDNW1A7RW18
I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing at all times.
I HAVE A DATE ON SUNDAY WITH WILL!!
XXXMobileMovieClub: To use your credit, click the WAP link in the next txt message or click here>> http://wap. xxxmobilemovieclub.com?n=QJKGIGHJGCB
Oh k...i'm watching here:)
Eh u remember how 2 spell his name... Yes i did. He v naughty make until i v wet.
Fine if thatâ€™s the way u feel. Thatâ€™s the way its gota b
England v Macedonia - dont miss the goals/team news. Txt ur national team to 87077 eg ENGLAND to 87077 Try:WALES, SCOTLAND 4txt/!â€1.20 POBOXox36504W45WQ 16+
Is that seriously how you spell his name?
I%œm going to try for 2 months ha ha only joking
So i _ pay first lar... Then when is da stock comin...
Aft i finish my lunch then i go str down lor. Ard 3 smth lor. U finish ur lunch already?
Fffffff. Alright no way I can meet up with you sooner?
< > spam +

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

LLM trained dataset:

```

() agent_training_dataset.json X
C:\Users\damini\Downloads\agent_training_dataset.json > {} 0
1  [
2  {
3    "input": "<p><b>HTML sample 0</b> Ceci est un texte d'exemple avec des erreurs pour décider quels agents exécuter.</p> Decide which agents to run",
4    "output": [
5      "html_strip_agent",
6      "spell_agent",
7      "sentence_split_agent",
8      "tokenizer_agent",
9      "lang_detect_agent",
10     "ner_agent",
11     "validation_agent"
12   ]
13 },
14 {
15   "input": "Este es un texto de ejemplo con errores para decidir qué agentes ejecutar. Decide which agents to run",
16   "output": [
17     "sentence_split_agent",
18     "tokenizer_agent",
19     "lang_detect_agent",
20     "ner_agent",
21     "validation_agent"
22   ]
23 },
24 {
25   "input": "இது உதாரண உரை. எந்த முகவர்கள் ஓடவேண்டும் என்பதை தீர்மானிக்க. Decide which agents to run",
26   "output": [
27     "spell_agent",
28     "sentence_split_agent",
29     "tokenizer_agent",
30     "lang_detect_agent",
31     "ner_agent",
32     "validation_agent"
33   ]
34 },
35 {
36   "input": "This is a sample text with mistakes to decide which agents to run. Decide which agents to run",
37   "output": [

```

IT22893666- Example Structured Dataset

	A	B	C	D	E	F	G	H
1	Customer	Gender	Age	Annual Income	Spending	Profession	Work Experience	Family Size
2	1	Male	19	15000	39	Healthcare	1	4
3	2	Male	21	35000	81	Engineer	3	3
4	3	Female	20	86000	6	Engineer	1	1
5	4	Female	23	59000	77	Lawyer	0	2
6	5	Female	31	38000	40	Entertainment	2	6
7	6	Female	22	58000	76	Artist	0	2
8	7	Female	35	31000	6	Healthcare	1	3
9	8	Female	23	84000	94	Healthcare	1	3
10	9	Male	64	97000	3	Engineer	0	3
11	10	Female	30	98000	72	Artist	1	4
12	11	Male	67	7000	14	Engineer	1	3
13	12	Female	35	93000	99	Healthcare	4	4
14	13	Female	58	80000	15	Executive	0	5
15	14	Female	24	91000	77	Lawyer	1	1
16	15	Male	37	19000	13	Doctor	0	1
17	16	Male	22	51000	79	Healthcare	1	2
18	17	Female	35	29000	35	Homemaker	9	5
19	18	Male	20	89000	66	Healthcare	1	6

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

NER Dataset

```
NER2 > {} dataset2.json > ...
1  [
2  {
3    "tokens": [
4      "Patient",
5      "name:",
6      "Kavindi",
7      "Wijesinghe",
8      "|",
9      "Condition:",
10     "Metacarpal",
11     "fracture",
12     "of",
13     "the",
14     "hand"
15   ],
16   "labels": [
17     "O",
18     "O",
19     "B-PERSON",
20     "I-PERSON",
21     "O",
22     "O",
23     "B-HEALTH_CONDITION",
24     "I-HEALTH_CONDITION",
25     "I-HEALTH_CONDITION",
26     "I-HEALTH_CONDITION",
27     "I-HEALTH_CONDITION"
28   ]
29 },
30 {
```

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

SLM Dataset

```
[
  {
    "input": [
      {
        "text": "Member States shall lay down the rules on other penalties applicable to infringement",
        "metadata": {
          "source_file": "GDPR.pdf",
          "page": 83,
          "chunk_id": "GDPR.pdf_p83_c4"
        }
      },
      {
        "text": "(70) Where personal data are processed for the purposes of direct marketing, the d",
        "metadata": {
          "source_file": "GDPR.pdf",
          "page": 13,
          "chunk_id": "GDPR.pdf_p13_c7"
        }
      },
      {
        "text": "Payment Card Industry Data Security Standard: Requirements and Testing Procedures, v4.",
        "metadata": {
          "source_file": "PCI-DSS.pdf",
          "page": 104,
          "chunk_id": "PCI-DSS.pdf_p104_c0"
        }
      }
    ],
    "output": {
      "entity": "DATE",
      "sensitivity_level": "Low",
      "reason": "The retrieved chunks discuss date as defined or regulated within the referenced standa",
      "regulation": "GDPR",
      "page": "83",
      "source_file": "GDPR.pdf"
    }
  },
  {
    "text": "Payment Card Industry Data Security Standard: Requirements and Testing Procedures, v4.",
    "metadata": {
      "source_file": "PCI-DSS.pdf",
      "page": 68,
      "chunk_id": "PCI-DSS.pdf_p68_c0"
    }
  },
  {
    "text": "For Compensating Controls Worksheet \u2013 Completed Example, correct wording at top c",
    "metadata": {
      "source_file": "PCI-DSS.pdf",
      "page": 2,
      "chunk_id": "PCI-DSS.pdf_p2_c1"
    }
  }
],
```

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

3.1. Suitability Analysis

3.1.1. Relevance to Individual Research Objectives:

	1	2	3	4
Data source 1	X			
Data source 2		X		
Data source 3		X		
Data source 4			X	
Data source 5				X
Data source 6				X
Data source 7			X	
Data source 8	X			

4. Methodology

4.1. Data Preprocessing:

Objective 1 : IT22893666

	Transformation technique							
	Data cleaning	Data Normalization	Data Standardization	Data Encoding	Handling Missing Data	Data Aggregation	Feature Engineering	Outlier Detection and Handling
Data source 1	X	X		X	X		X	X
Data source 8	X	X		X	X		X	X

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Objective 2: IT22556684

	Transformation technique							
	Data cleaning	Data Normalization	Data Standardization	Data Encoding	Handling Missing Data	Data Aggregation	Feature Engineering	Outlier Detection and Handling
Synthetic (NER dataset)	X				X	X	X	
Synthetic (SLM)	X					X	X	

Objective 3: IT2254931

	Transformation technique							
	Data cleaning	Data Normalization	Data Standardization	Data Encoding	Handling Missing Data	Data Aggregation	Feature Engineering	Outlier Detection and Handling
Data source 4	X		X	X		X	X	
Data source 7	X			X	X			

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Objective 4: IT22204752

	Transformation technique							
	Data cleaning	Data Normalization	Data Standardization	Data Encoding	Handling Missing Data	Data Aggregation	Feature Engineering	Outlier Detection and Handling
Data source 1		X	X	X		X	X	
Data source 8						X	X	

4.2. Scalability

- Data Source 1:

Synthetic sensor- and transaction-like data streams were generated with configurable volumes to evaluate real-time data quality assessment. The dataset includes injected missing values, duplicates, and anomalies, enabling scalable testing of explainable preprocessing, benchmarking, and rollback mechanisms. Dataset sizes range from 10,000 to 1 million records, allowing the framework to be evaluated under varying data loads without dependency on a fixed dataset.

- Data Source 2:

A synthetic text dataset was created to train and evaluate the NER model used in privacy preservation. The dataset simulates realistic, sensitive, and non-sensitive entities without using real personal data. All instances are annotated using the BIO tagging scheme and stored in JSON format. The dataset contains approximately 10,000 labeled samples, ensuring sufficient coverage while maintaining computational efficiency.

- Data Source 3:

A regulatory knowledge dataset was generated based on privacy standards such as GDPR, PDPA, and PCI. The data is structured in a question-and-answer (Q&A) format and stored in JSON for

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

consistency and ease of integration. With approximately 400–500 entries, this dataset supports scalable rule interpretation and compliance-related reasoning within the framework.

- Data Source 4:

A large synthetic, unstructured text dataset was created to train the LLM responsible for autonomous agent selection and orchestration within the multi-agent preprocessing framework. The dataset comprises approximately 50,000 text records containing systematically injected quality issues, including spelling errors, grammatical noise, duplicated content, mixed-language text, and malformed sentences. By providing explicit mappings between text characteristics and required preprocessing actions, this dataset enables the LLM to learn scalable and context-aware decision-making for selecting appropriate preprocessing agents in real-time unstructured data streams.

- Data Source 5:

Multiple versions of a structured dataset were synthetically generated to simulate schema drift scenarios such as column addition, removal, renaming, and data type changes. These versioned datasets enable systematic evaluation of real-time schema drift detection and automated handling under controlled and reproducible conditions.

- Data Source 6:

Public Kaggle datasets were used to simulate schema drift by intentionally modifying schemas through column addition, removal, and renaming. These datasets allow evaluation of the schema drift framework on realistic real-world data structures, supporting scalability beyond synthetic-only scenarios.

- Data Source 7:

Real-world structured datasets obtained from Kaggle were used to validate the framework under realistic data distributions, noise, and attribute diversity. These datasets include mixed numerical and categorical fields with missing values and inconsistent formatting, enabling scalable testing of structured data quality mechanisms.

- Data Source 8:

Public Kaggle unstructured datasets, such as text or log-style data, were used to evaluate multi-agent preprocessing, quality scoring, and integrity checks in simulated streaming environments. Dataset sizes vary depending on the selected source, demonstrating that the framework can operate on diverse unstructured data at scale.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

4.3. Feature extraction

IT22893666

Objective: identify the most relevant features for each target column requiring prediction.

- Dependency analysis using Mutual Information, Chi-Square tests, and Spearman correlation to measure feature-target relationships
- Top 3 feature selection based on highest dependency scores for each target variable
- Computational efficiency through dimensionality reduction from full feature space to three predictors
- Overfitting prevention by eliminating irrelevant and weakly correlated features
- Enhanced interpretability for SHAP explanations through focused feature space

Objective 4: IT22204752

Preprocessing

Objective: Clean column text for reliable comparison.

- Lowercased names, removed symbols/spaces, normalized separators.
- Removed nulls when collecting sample values.

Feature Representation

Objective: Convert column info into numeric features for ML.

- Built column descriptions (name + type + sample values).
- Generated embeddings (SentenceTransformer / TF-IDF fallback).
- Extracted column features: null rate, unique ratio, patterns, numeric stats.
- Computed pairwise features: embedding cosine, name similarity, type match, value overlap.

Label Mapping

Objective: Convert ground truth rename mapping into labels.

- Rename = 1
- Not rename = 0

Feature Selection / Evaluation

Objective: Check which features improve accuracy.

- Compared: name only, embedding only, name+embedding, +column features.
- Ranked important features using model weights.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

5. Modelling and Results

The results section should present the key findings derived from analyzing the data. It should answer the research questions or hypotheses posed at the beginning of the study, based on the methods and techniques applied during the analysis.

5.1. Key Insights:

- **Structured Data Quality :**

The real-time quality assessment framework effectively identified common data quality issues such as missing values, duplicates, and outliers across both synthetic and Kaggle-based structured datasets. Quality degradation patterns were observable under increased data velocity and volume, particularly in streams with injected anomalies. Explainable preprocessing decisions improved transparency by clearly linking detected issues to applied corrections. Benchmarking results showed that adaptive preprocessing maintained stable quality scores over time, while the rollback mechanism successfully restored previous stable states when quality thresholds were violated, preventing downstream data corruption.

- **Privacy Preservation and Sensitive Data Detection :**

The NER-based privacy mechanism demonstrated reliable detection and classification of sensitive and non-sensitive entities in streaming text data. BIO-tagged synthetic datasets enabled consistent learning of entity boundaries without exposure to real personal information. The integration of regulatory knowledge from GDPR, PDPA, and PCI supported context-aware sensitivity classification, reducing false positives in non-sensitive fields. Real-time masking and transformation actions were applied with minimal latency, indicating feasibility for deployment in high-throughput streaming pipelines.

- **Unstructured Data Quality and Multi-Agent Processing :**

Analysis of synthetic and Kaggle unstructured datasets revealed recurring quality issues such as language inconsistencies, malformed text, duplication, and semantic noise. The agentic framework successfully orchestrated specialized preprocessing agents based on detected text characteristics. Visual inspection of quality scores before and after preprocessing showed consistent improvements in readability, structural consistency, and semantic integrity. The multi-agent approach scaled effectively with increasing data volume, maintaining stable processing latency while improving overall data reliability in simulated streaming conditions.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

- **Schema Drift Detection and Management :**

The schema drift detection mechanism accurately identified structural changes including column addition, removal, renaming, and data type modifications across both synthetic schema versions and modified Kaggle datasets. Drift events were detected in real time before downstream failures occurred, enabling proactive handling. Rule-based classification of drift types supported appropriate automated responses, such as schema mapping updates or validation alerts. Controlled experiments with versioned datasets demonstrated that the framework could adapt to evolving schemas without interrupting streaming data flows.

- **Cross-Objective Observations:**

Across all objectives, the unified autonomous framework reduced manual intervention by integrating quality assessment, privacy enforcement, unstructured data handling, and schema drift management within a single streaming pipeline. Experiments showed that early-stage intervention in the pipeline significantly minimized error propagation and compliance risks. The use of verified synthetic data alongside public datasets ensured reproducibility while maintaining realism, demonstrating the framework's robustness across diverse data types and operational scenarios.

5.2. Challenges Faced During Data Analysis:

- **Simulating Real-Time Streams:** Difficulty in replicating the high velocity, irregularity, and burstiness of real-world streaming data (e.g., sensor/log streams) using static or purely synthetic data.
- **Balancing Control and Realism:** Ensuring the synthetic datasets (for quality, privacy, and schema drift) are both perfectly controlled (to isolate variables) and realistically representative of industry data.
- **Computational Scalability:** Ensuring the LLM-based multi-agent orchestrator for unstructured data (Objective 3) operates with low enough latency to be practical in a high-speed streaming environment.
- **Regulatory Knowledge Extraction:** The complexity of using LLMs to accurately convert ambiguous, unstructured legal documents (GDPR/PCI) into machine-executable privacy constraints for the Small Language Model (SLM).
- **Generalization across Diverse Data:** Avoiding models that overfit to the training data and ensuring the autonomous framework remains effective and reliable when applied to diverse, unseen Kaggle datasets (Source 6, 7, 8) with varied structures and noise patterns.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

6. References

- [1] <https://www.nhsinform.scot/illnesses-and-conditions/a-to-z/>
- [2] <https://www.kaggle.com/datasets/bhavikbb/password-strength-classifier-dataset>
- [3] <https://www.kaggle.com/datasets/fenilsonani/email-data-for-email-classification>
- [4] <https://forebears.io/sri-lanka/surnames>
- [5] https://en.wikipedia.org/wiki/Category:Surnames_of_Sinhalese_origin